

EEE 491 Lab Assignment

Lab-MEL: MEL Filter Banks

1. Technical Specifications

Lab-MEL shall read data from Lab-FFT's dual-port RAM. At least 32 MEL filters shall be applied for the frequency data ranging from about 100Hz to about 4 KHz. Each triangular shaped filter covers a band according to the MEL frequencies. A number of data at each filter output shall be used to calculate the band energy. Band energies shall be stored in a simple dual-port RAM with 32 words. Data width of the RAM shall be at least 16-bit. Use "Block Memory generator" and "Multipliers" from the Vivado "IP Catalog".

Lab-MEL in FPGA shall have the following input/output (IO) signals.

- "reset_in" input control signal, which resets all the registers in the design.
- "clock_in" input clock signal of the 100MHz clock oscillator on the Basys-3 board.
- "fft_addr_out" output signal that addresses the read-only port of the dual-port RAM in Lab-FFT.
- "fft_data_in" input signal from the read-only port of the dual-port RAM in Lab-FFT.
- "mem_addr_in" input signal that addresses the read-only port of the dual-port RAM in Lab-MEL.
- "mem_data_out" output signal of the read-only port of the dual-port RAM in Lab-MEL.
- "start_in" input control signal which starts the MEL filtering function for all FFT output data. This signal is active high asserted on a single clock pulse (refer to waveform in Lab-ADC assignment).
- "ready_out" output status signal that indicates if MEL filtering function is completed over the FFT data, output is written to dual-port RAM of Lab-MEL and the function is ready for the next execution. This signal is active high and asserted low just after the start signal is asserted (refer to waveform in Lab-ADC assignment).

After verifying your Lab-MEL design by using your test-bench simulation results, integrate Lab-MEL with the Lab-PCB, ADC board, Lab-ADC, Lab-WINDOW, Lab-CTRL, Lab-FFT, and Lab-DEBUG. Connect Lab-DEBUG to Lab-MEL output. Control and observe the process sequence using start_in signal of Lab-CTRL and ready_out signals of the components. On the Basys-3 board, use a button for start_in signaling and a LED for each ready_out signals of the components.

2. Demonstration

- Present your test bench simulation results: inputs, outputs (waveforms). Test bench generates constant value test data as FFT output, starts the Lab-MEL, and waits for the ready signal and then it reads data from the dual-port RAM of Lab-MEL. Show that the results satisfy the MEL filter function.
- Configure your integrated design on Basys-3 board. Use start_in button and ready_out LEDs of the components to observe/verify process sequence. If you need you may add more LEDs for further observations.
- Prepare the Lab-CTRL demonstration set up. Run the integrated system and transfer the MEL output data of each frame to MATLAB. Compare and verify MEL filter results with the Lab-MATLAB results for sinus input frequencies: 100Hz, 500Hz, 1KHz, 4KHz with 1.5V peak-to-peak amplitude for each frequency.

3. Guidance

- You can refer to <https://link.springer.com/content/pdf/bbm:978-3-319-49220-9/1.pdf>, and as an example https://isip.piconepress.com/courses/msstate/ece_8463/lectures/current/lecture_12/lecture_12_08.html
- Implement the MEL filter banks as modeled in Lab-MATLAB assignment on the FPGA.